Phase diagram

By Galib93

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Phase

- A phase may be defined as : any homogeneous part of a system having all physical and chemical properties the same throughout system. A system may consist of one or more than one phases.
- 1) A system containing only liquid water, it contains one-phase.
- 2) A system containing liquid water and water vapour, it contains two-phase.
- 3) A system containing liquid water, water vapour and solid ice, it contains three-phase.
- **A system consisting of one phase only is called a homogeneous system.**
- **A system consisting of two or more phases is called a heterogeneous system.**

Components

- The term **component** may be defined as : the least number of independent chemical constituents in terms of which the composition of every phase can be expressed by means of a chemical equation.
- **Explanation**
- To understand the above definition and to use it for finding the number of components of a system, we need to remember that :
- *a*) The chemical formula representing the composition of a phase is written on LHS.
- *b*) The rest of the chemical constituents existing independently in the system as represented by chemical formulas are placed on RHS.
- *c*) The quantities of constituents on RHS can be made minus (–) or zero (0) to get at the composition of the phase on LHS.

- **1) Pure substances:**
- A pure substance made of one chemical species only, is considered as one phase.
- Thus oxygen, benzene, and ice are all 1-phase systems.
- It must be remembered that a phase may or may not be continuous.
- Thus, whether ice is present in one block or many pieces, it is considered one phase.
- **2) Mixtures of gases:**
- All gases mix freely to form homogeneous mixtures.
- Therefore any mixture of gases, say O_2 and N_2, is a 1-phase system.
- **3) Miscible liquids:**
- Two completely miscible liquids yield a uniform solution.
- Thus a solution of ethanol and water is a 1-phase system.

- **4) Non-miscible liquids:**
- A mixture of two non-miscible liquids on standing forms two separate layers.
- Hence a mixture of chloroform ($CHCl_3$) and water constitutes a 2-phase system.
- **5) Mixtures of solids:**
- By definition, **a phase must have throughout the same physical and chemical properties.**
- Ordinary sulphur as it occurs in nature is a mixture of monoclinic and rhombic sulphur.
- These allotropes of sulphur consist of the same chemical species but differ in physical properties.
- Thus mixture of two allotropes is a 2-phase system.

- **1) Water and sulphur are 1-component systems**
- **Water system has three phases:** *ice*, *water*, and *water vapour*.
- The composition of all the three phases is expressed in terms of one chemical individual H_2O.
- Thus water system has one component only.
- **Sulphur system has four phases :** rhombic sulphur, monoclinic sulphur, liquid sulphur and sulphur vapour.
- The composition of all these phases can be expressed by one chemical individual sulphur (S).
- Hence it is a 1-component system.
- Now it is clear that, when all the phases of a system can be expressed in terms of one chemical individual, it is designated as 1-component system or one-component.

- **2) Mixture of gases**
- A mixture of gases, say O_2 and N_2, constitutes one phase only.
- Its composition can be expressed by two chemical substances O_2 and N_2.
- Hence a mixture of O_2 and N_2 has two components.
- In general the number of components of a gaseous mixture is given by the number of individual gases present.

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- **3) Sodium chloride solution**
- A solution of sodium chloride in water is a 1-phase system.
- Its composition can be expressed in terms of two chemical individuals, sodium chloride and water.
- Therefore an aqueous solution of sodium chloride or any other solute is a two-component or 2-component system.

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Degrees of freedom

- The term **degree of freedom** represented by F in the phase rule equation ($F = C - P + 2$).
- It is defined as follows : the least number of variable factors (concentration, pressure and temperature) which must be specified so that the remaining variables are fixed automatically and the system is completely defined
- A system with $F = 0$ is known as non-variant or having no degree of freedom.
- A system with $F = 1$ is known as univariant or having one degree of freedom.
- A system with $F = 2$ is known as bivariant or having two degrees of freedom.

Degrees of freedom

- **1) For ice-water-vapour system, *F* = 0:**
- In the system ice ↔ water ↔ vapour, the three phases coexist at the freezing point of water.
- Since the freezing temperature of water has a fixed value,
- the vapour pressure of water has also a definite value.
- The system has two variables (temperature and
- pressure) and both these are already fixed.
- Thus the system is completely defined automatically,
- there being no need to specify any variable. Hence **it has no degree of freedom (*F* = *0*).**

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Degrees of freedom

- **2) For water ↔ water vapour, *F* = 1:**
- The system water in equilibrium with water vapour, has two variables *temperature* and *pressure*.
- At a definite temperature the vapour pressure of water can have only one fixed value.
- Thus if one variable (temperature or pressure) is specified, the other is fixed automatically.
- Hence the system water has **one degree of freedom (*F* = *1*).**

- **3) For a pure gas, *F* = 2:**
- For a given sample of any pure gas $PV = RT$.
- If the values of pressure (P) and temperature (T) are specified, volume (V) can have only one definite value, or that the volume (the third variable) is fixed automatically.
- Hence a system containing a pure gas has **two degrees of freedom (*F* = 2).**
- **4) For a mixture of gases, *F* = 3:**
- A system containing a mixture of two or more gases is completely defined when its composition, temperature and pressure are specified.
- If pressure and temperature only are specified, the third variable *i.e.*, composition could be varied. Since it is necessary to specify three variables to define the system completely, a mixture of gases has **three degrees of**
- **freedom (*F* = 3).**

Phase rule

- The phase rule is an important generalization dealing with the behaviour of heterogeneous systems.
- In general it may be said that with the application of *phase rule* it is possible to predict qualitatively by means of a diagram the effect of changing pressure, temperature and concentration on a heterogeneous system in equilibrium.
- It may be stated mathematically as follows :

$$F = C - P + 2$$

- where F is the number of degrees of freedom, C is the number of components and P is the number of phases of the system.

- **For a one-component system** we can write the phase rule equation as :

$$F = C - P + 2 = 1 - P + 2 = 3 - P$$

- Three cases may arise :
- **Case 1:** When only **one phase** is present,

$$F = 3 - 1 = 2$$

- Thus the system is **bivariant.**
- It can be completely defined by specifying the two variables, temperature and pressure.
- Or that, both the temperature and pressure can be varied independently.
- Therefore **a single phase is represented by an area on P, T-graph.**

Phase rule

- **Case 2:**
- When **two phases** are in equilibrium,
- $F = 3 - 2 = 1$
- The system then has one degree of freedom and is termed **monovariant.**
- This means that the pressure cannot be changed independently if we change the temperature.
- The pressure is fixed automatically for a given temperature.
- **A two-phase system is depicted by a line on a P, T-graph.**

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Phase rule

- **Case 3:** When **three phases** are in equilibrium,

  $F = 3 - P = 3 - 3 = 0$

- The system has zero degree of freedom and is termed **nonvariant or invariant.**
- This special condition can be attained at a definite temperature and pressure.
- The system is, therefore, defined completely and no further statement of external conditions is necessary.
- **A three-phase system is depicted by a point on the P, T-graph.**
- At this point the three phases (solid, liquid, vapour) are in equilibrium and, therefore, it is referred to as the **Triple point.**

Phase diagram of two components system

- A phase diagram of a two components system is a plot showing the conditions of temperature and compositions under which two or more physical states can exist together in a state of dynamic equilibrium.
- A typical phase diagram for a two-components system consists of :
- a) the regions or areas
- b) the lines or curves and
- c) the eutectic point.

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Phase diagram of two components system

- When a single phase is present in a two-component system, the degree of freedom is three,

  $F = 2 - 1 + 2 = 3$

- This means that three variables must be specified in order to describe the condition of the phase.
- Thus in such a system, in addition to pressure and temperature the concentration of one of the components has also to be given.
- For graphic representation of these variables, three coordinate axes at right angles to each other would be required.
- Therefore the phase diagram obtained would be a solid model.

Phase diagram of two components system

- For the sake of having simple plane diagrams we generally consider only two variables, the third one being a constant.
- For example, for a **solid/liquid** equilibrium, the gas phase in usually absent and the effect of pressure on the equilibrium is very small.
- Thus when a two-component system consists of solid and liquid phases only, the effect of pressure may be ignored.
- Then it is necessary to take into account the remaining variables *viz.*, temperature and concentration.
- Such a solid/liquid system with the gas phase absent is called a **condensed system.**

Reduced phase rule

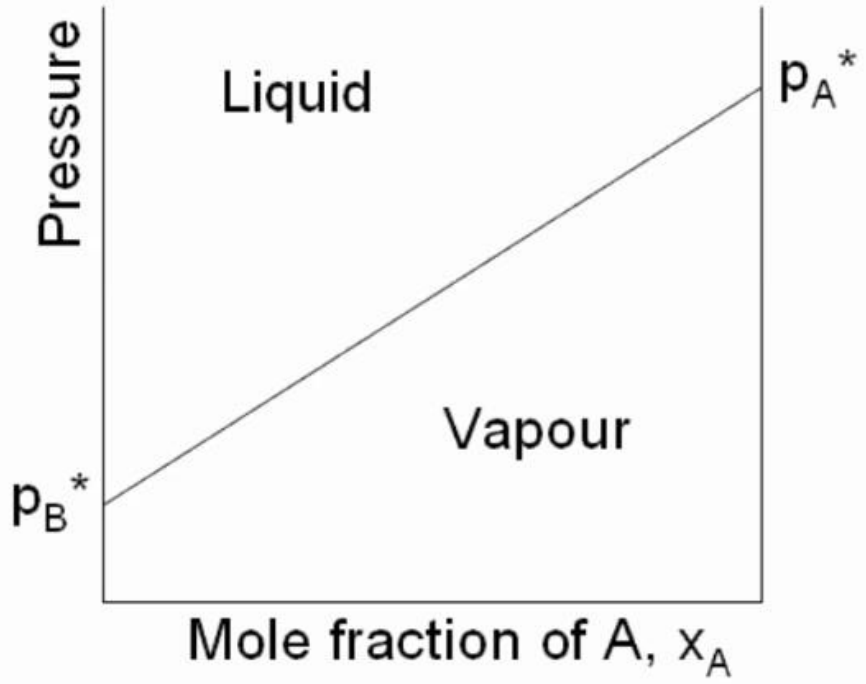
- The experimental measurements of temperature and concentration in condensed systems are usually carried out under atmospheric pressure.
- Since the degree of freedom in such a case is reduced by one, we may write the **reduced phase rule** as

$$F' = C - P + 1$$

- where F' gives the remaining degrees of freedom of the system.
- The reduced phase rule is more convenient to apply to solid/liquid two-component condensed system.
- Since the only variables for two-component solid/liquid systems are temperature and composition, the phase diagrams for such systems consist of temperature-concentration graphs (*TC graphs*).

Vapour-pressure diagrams

- The partial vapour pressure of the components of an ideal solution of two volatile liquids are related to the composition of the liquid mixture by Raoult's law
- $p_A = x_A p^*_A$ and $p_B = x_B p^*_B$
- Where p^*_A and p^*_B are the vapour pressure of pure A and pure B respectively. The total vapour pressure of the mixture is
- $P = p_A + p_B = x_A p^*_A + x_B p^*_B = p^*_B + (p^*_A - p^*_B)x_A$
- This expression shows that the total vapour pressure changes linearly with the composition from p^*_B to p^*_A as x_A changes from 0 to 1.

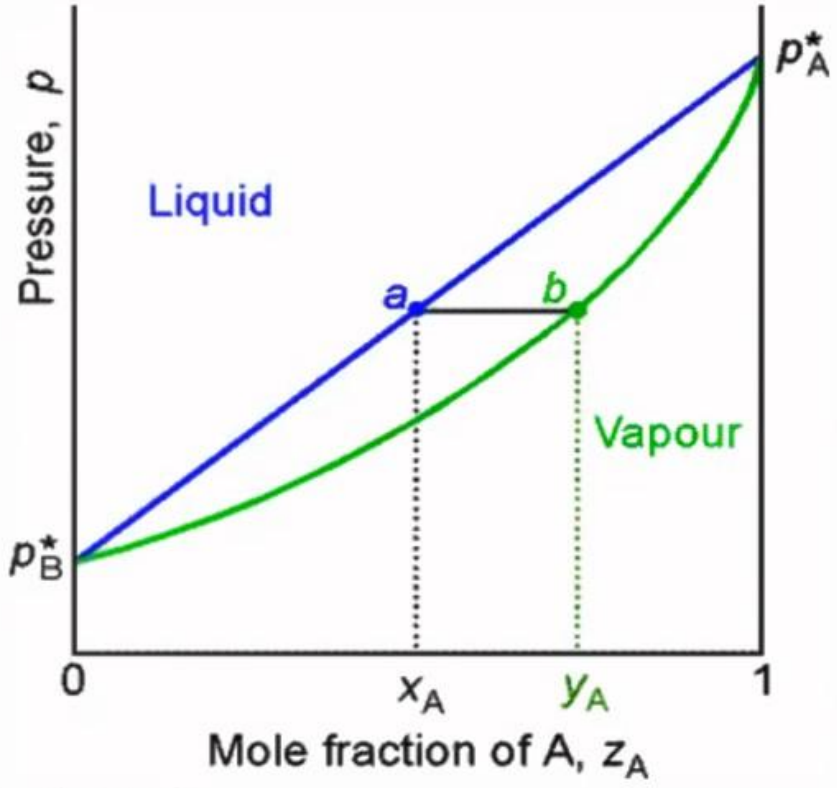


The variation total vapour pressure of a binary mixture with the mole fraction of A in the liquid when Raoult's law is obeyed.

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Isopleths, tie line

- For distillation, both vapor and liquid compositions are of interest. Thus we combine the liquid composition diagram and vapor composition diagram into one.

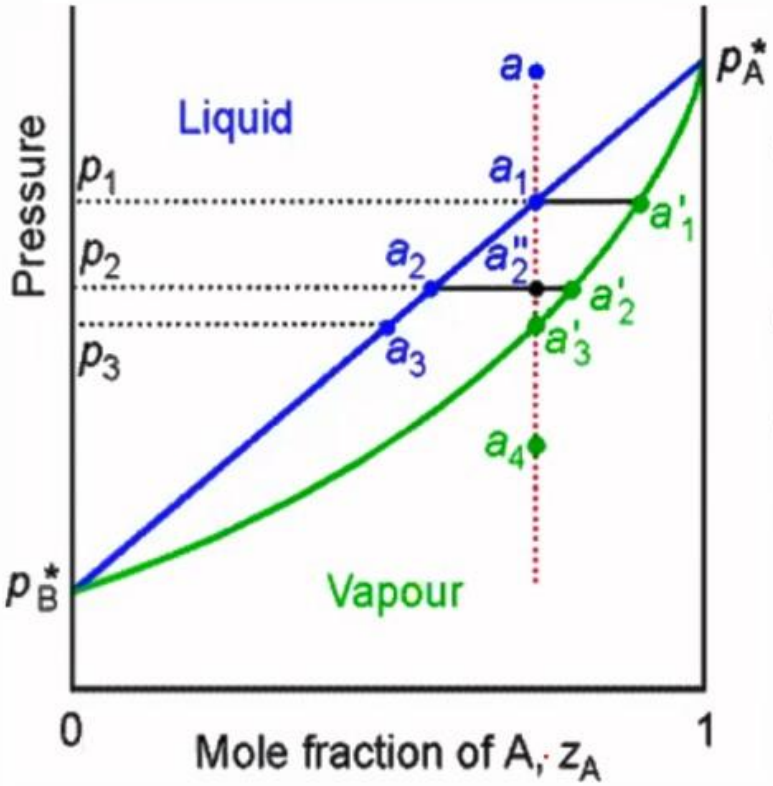


The dependence of the total vapour pressure of an ideal solution on the mole fraction of A in the entire system. A point between the two lines corresponds to both liquid and vapour being present (P=2;F=1); outside that region there is only one phase present (P=1;F=2). The mole fraction of A is denoted z_A, as explained in the text.

Isopleths, tie line

- The point a indicates the vapor pressure of a mixture of composition x_A and b indicates the composition of the vapor that is in equilibrium with the liquid at that pressure.
- Note that when two phases are in equilibrium, P = 2, so F' = 1. Thus if composition is specified, the pressure at which the two phases are in equilibrium is fixed.

Isopleths, tie line



The points of the pressure-composition diagram discussed in the text. The vertical line through *a* is an isopleth, a line of constant composition of the entire system.

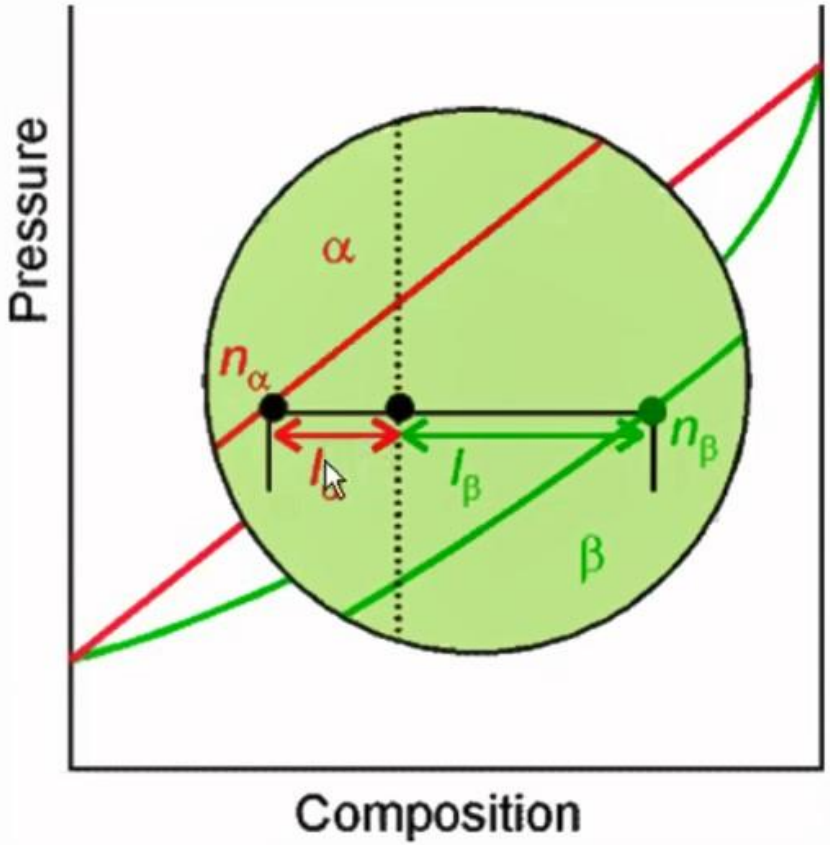
- As we decrease the pressure we travel down the line a (constant composition vertical line) to a4.
- **This vertical line is called an isopleth.**
- at p_1 we have liquid with composition a_1 with very little vapor at composition a_1'.

- At a_1 the liquid can exist in equilibrium with its vapour, a'_1.
- **A line joining two points representing phases in equilibrium is called tie line.**
- At p_2 we have liquid with composition and vapor with composition a in equilibrium (the overall composition of the system is a_2").
- Note also that the two phases are in equilibrium and F' = 1 for all points between the two lines.
- Hence for a given pressure (such as) the variance is zero, and the vapor and liquid phases have fixed compositions.
- At p_3 we have very little liquid with composition a_3 in equilibrium with mostly vapor at composition a_3. At a_4 we have pure vapor only.

Lever rule

- A point in the two-phase region of a phase diagram indicates not only qualitatively that the liquid and vapor are present, but represents quantitatively the relative amounts of each.
- The lever rule is a rule used to determine the mole fraction of each phase of a binary equilibrium phase diagram.
- It can be used to determine the fraction of liquid and solid phases for a given binary composition and temperature.
- To find the relative amounts of two phases α & β in equilibrium, we measure distances on the tie line, l_α and l_β between the two phases and use **the lever rule:**
- $$n_\alpha l_\alpha = n_\beta l_\beta$$
- where n_α is the amount of phase α and n_β is the amount of phase β.

Lever rule



The lever rule. The distances l_α and l_β are used to find the proportions of the amounts of phases α (such as vapour) and β (for example, liquid) present at equilibrium.

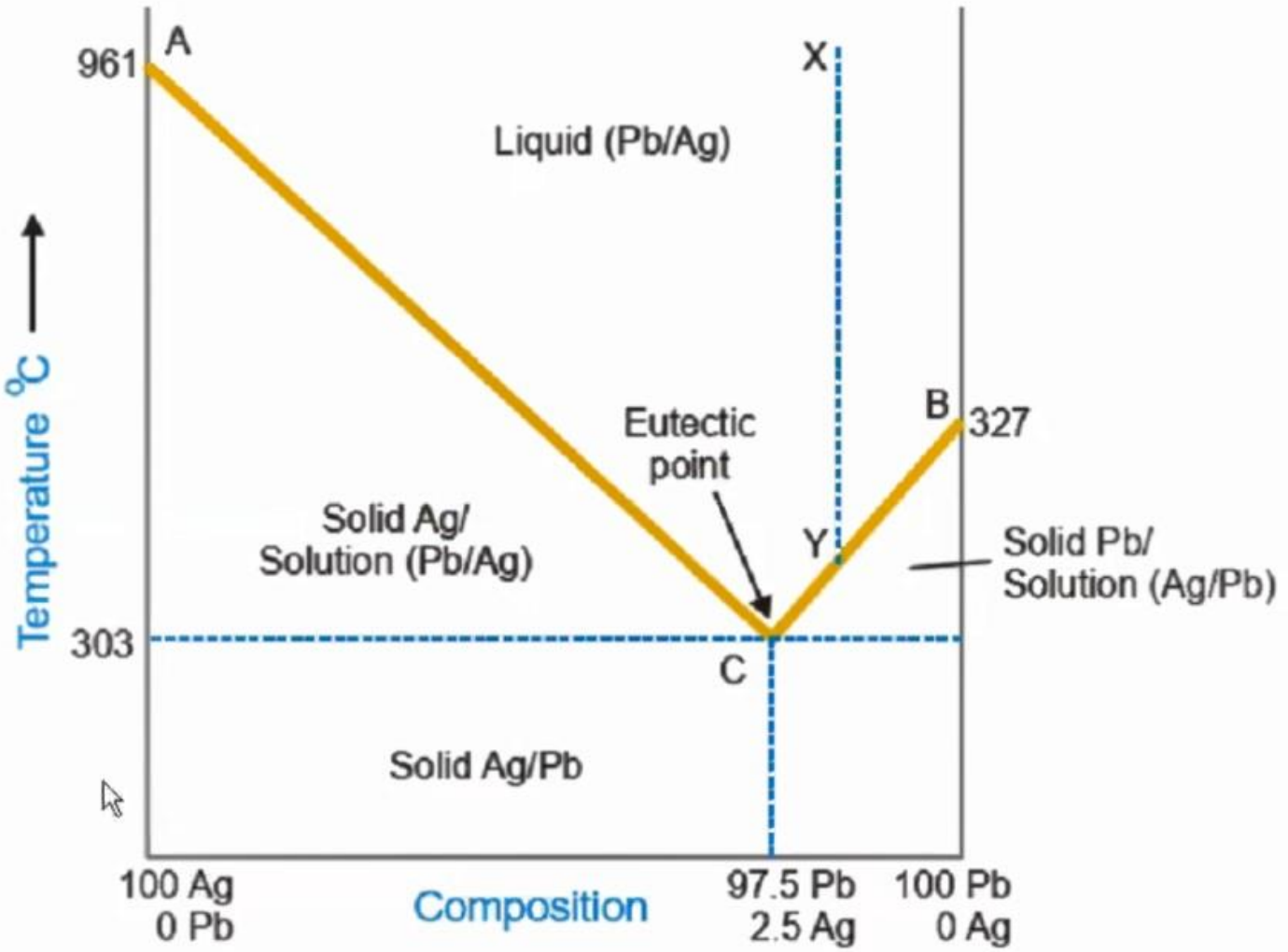
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Temperature-composition diagrams

Phase diagram of silver-lead system

- This system has **two components** and **four phases**.
- The phases are :
- i) solid silver
- ii) solid lead
- iii) solution of molten silver and lead and
- iv) vapour.
- The boiling points of silver and lead being considerably high, the vapour phase is practically absent.
- Thus Ag/Pb is a condensed system with three phases. In such a case, pressure can have no effect on the system.
- Therefore we need consider only the two remaining variables, namely the temperature (T) and concentration (C). The complete *TC diagram* of the system Ag/Pb is shown in figure.

Temperature composition diagrams



- **The main features of the diagram are :**
- a) Two curves, AC and BC
- b) Eutectic point, C
- c) Three areas : i) above ACB, ii) below AC,
- iii) below BC
- **Curve AC is** the freezing point curve of Ag. *A* represents the freezing point or melting point of solid silver (961ºC) and the curve AC shows that the addition of lead decreases the melting point along it.
- The phases in equilibrium along AC are solid silver and solution of silver and lead.
- Applying the reduced phase rule equation
- $F' = C - P + 1 = 2 - 2 + 1 = 1$
- Thus the system Ag/solution is monovariant.

- **Curve BC:** It is the Freezing point curve of Pb.
- B represents the melting point of solid lead (327°C) and the curve BC shows that the melting point is lowered by addition of silver.
- The phases in equilibrium along BC are solid lead and solution. The system is monovariant.
- **The Eutectic point C:** The curves AC and BC intersect at C, which is called the eutectic point.
- Here three phases solid Ag, solid Pb, and solution are in equilibrium.
- Applying the reduced phase rule equation
- $F' = C - P + 1 = 2 - 3 + 1 = 0$

- Thus the system Ag/Pb/solution at C is nonvariant.
- Both the variables, temperature (303ºC) and composition (97.5% Pb, 2.5% Ag) are fixed.
- If we raise the temperature above the eutectic temperature, the solid phases Ag and Pb disappear and if you cool below it, you will land in the solid Ag/Pb area where solution phase is nonexistent.
- **The Area above ABC:** This region represents the single phase system, the solution of molten Ag and Pb.
- Applying the reduced phase rule equation, we have

  $$F' = C - P + 1 = 2 - 1 + 1 = 2$$

- Thus the system solution Ag/Pb is bivariant.

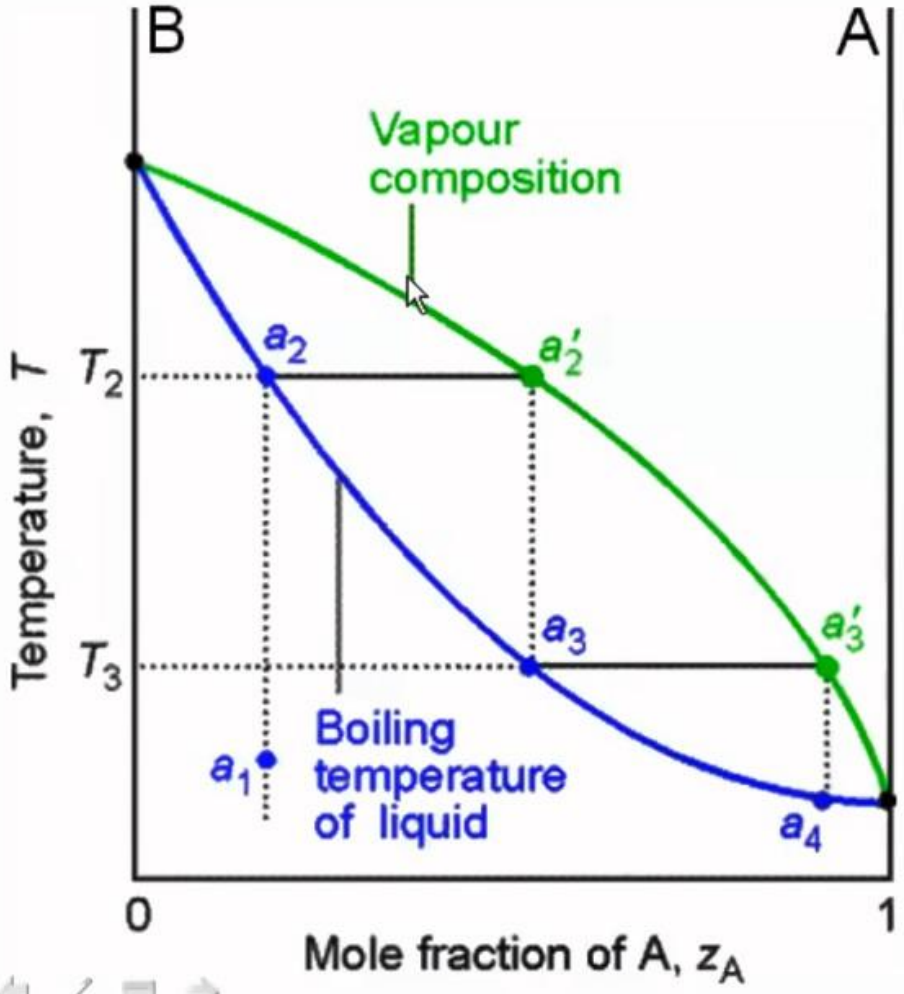
- **The area below AC** represents the phases Ag + solution, while that below BC the phases Pb + solution.
- The area below the temperature 303°C, represents solid Ag + solid Pb.
- All these areas have two phases and one degree of freedom,

$$F = C - P + 1 = 2 - 2 + 1 = 1$$

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Distillation of mixtures

- To discuss distillation, we need a temperature composition diagram (pressure is held constant).



The temperature--composition diagram corresponding to an ideal mixture with the component **A** more volatile than component **B**. Successive boilings and condensations of a liquid originally of composition a_1 lead to a condensate that is pure **A**. The separation technique is called fractional distillation.

Distillation of mixtures

- The region between the lines in the above figure is a two-phase region with F' = 1 and hence at a given T, the composition of the phases in equilibrium are fixed.
- As we heat the liquid with composition from a_1, it will start to boil when it reaches T_2.
- The vapor will be richer in the more volatile component (A) and will have composition a'_2.
- In a simple distillation the vapor is withdrawn and condensed.
- If the vapor is completely withdrawn and condensed the first drop gives a liquid of composition a_3, which is richer in the more volatile component, A, than the original liquid.

Distillation of mixtures

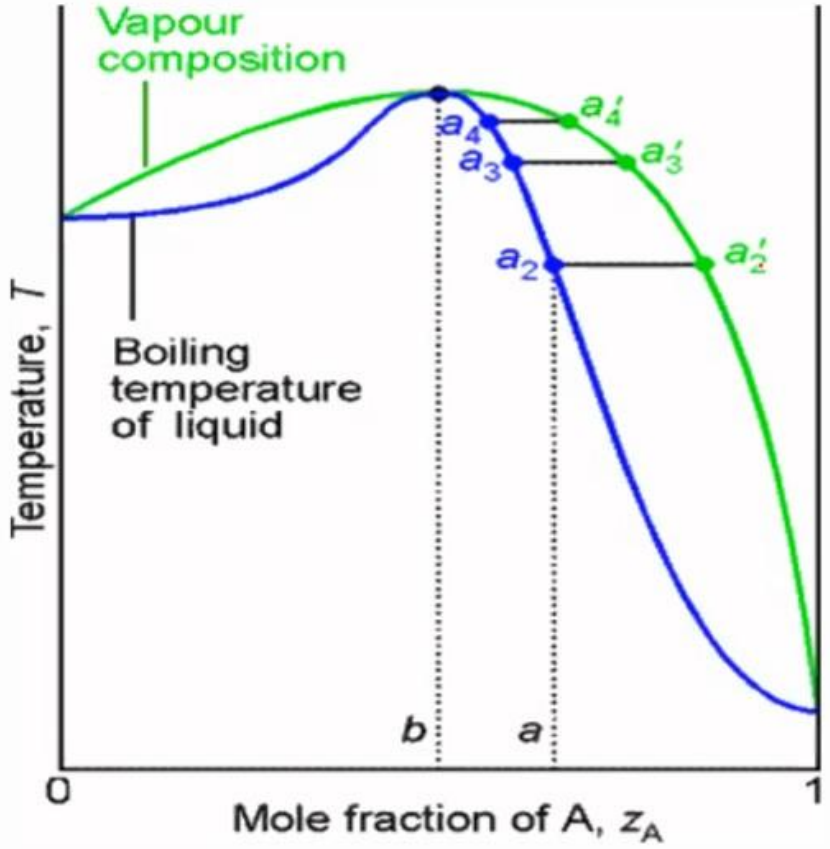
- In **Fractional distillation**, the boiling and condensation cycle is repeated successively.
- We can follow the next change by examining what happens when the condensate of composition a_3 is reheated.
- The mixture will now boil at T_3 and the composition of the vapor will be a'_3. Then we go to a_4 etc.

Azeotropes

- A maximum in a phase diagram may occur when favorable interactions between A and B molecules reduce the vapor pressure of the mixture below the ideal value.
- In such cases the excess Gibbs energy (G^E) is negative so the mixing is favorable and the liquids are miscible.
- Examples are trichloromethane/acetone and nitric acid/water mixtures.

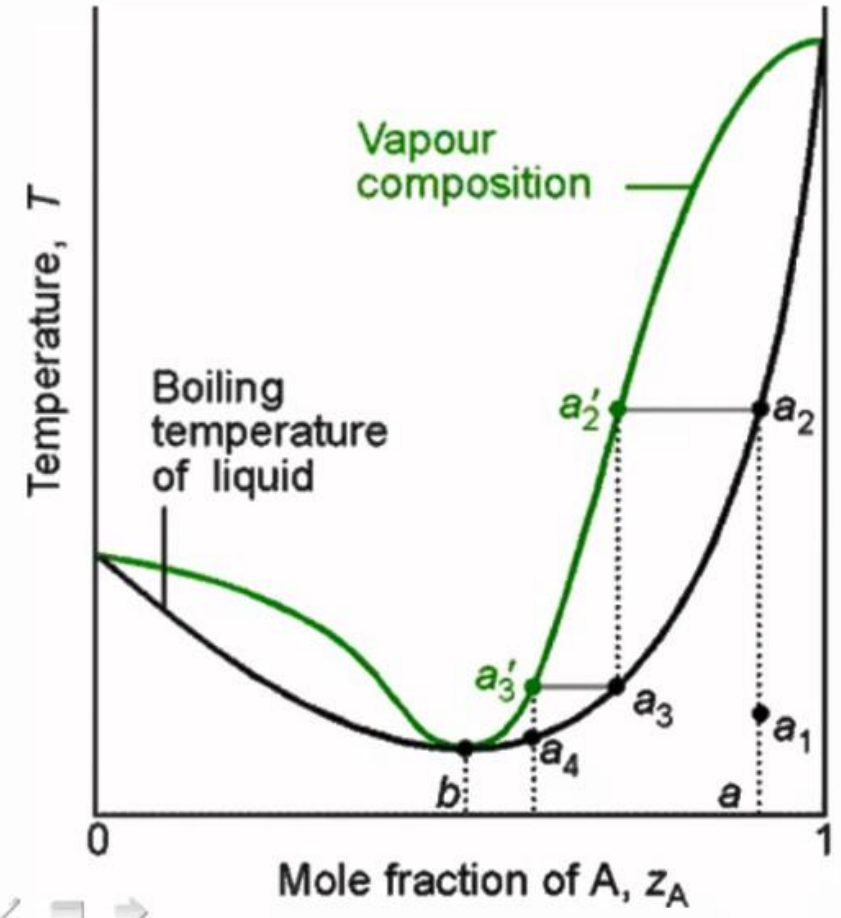
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Azeotropes



A high-boiling azeotrope. When the liquid of composition *a* is distilled, the composition of the remaining liquid changes towards *b* but no further.

- Phase diagrams showing a minimum indicate that the mixture is destabilized relative to the ideal solution, the A-B interactions then being unfavorable.
- For such mixtures G^E is positive (less favorable to mixing than ideal). Examples are dioxane/water and ethanol water mixtures.



A low-boiling azeotrope. When the mixture at *a* is fractionally distilled, the vapour in equilibrium in the fractionating column moves towards *b* and then remains unchanged.

Azeotropes

- Consider a liquid composition of a in the **high boiling azeotrope.**
- The vapor (at a'_2) of the boiling mixture (at a_2) is richer in A.
- If that vapor is removed (and condensed elsewhere) the remaining liquid will move to a composition that is richer in B, such as that represented by a_3, and the vapor in equilibrium with this mixture will have composition a'_3.
- The boiling point of the liquid drops and the vapor becomes richer in B.

Azeotropes

- When the remaining liquid reaches composition b, the vapor has the same composition as the liquid.
- Evaporation occurs without change of composition.
- The mixture is said to form an **azeotrope**.
- When the azeotrope is reached the two liquids cannot be separated.

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Liquid-liquid phase diagrams

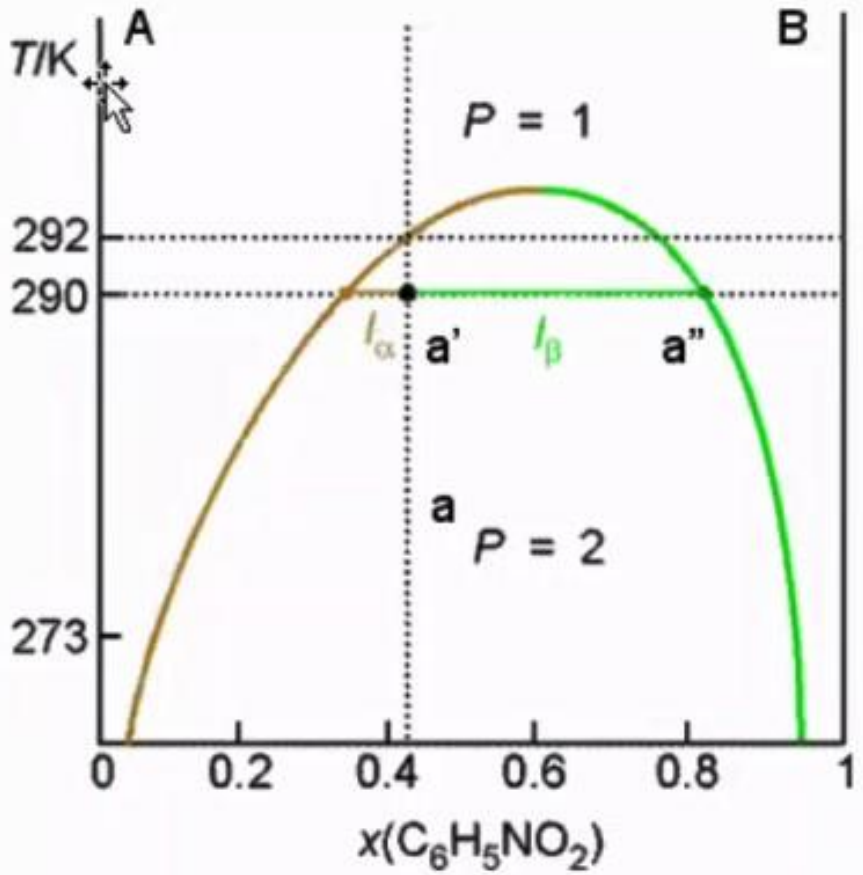
- We will study temperature-composition diagrams for systems that consist of pairs of **partially miscible liquids**, (liquids that do not mix in all proportions at all temperatures).
- An example is hexane and nitrobenzene.
- When P = 2, F' = 1, and fixed T will determine compositions of the immiscible liquid phases.

Phase separation

- Suppose small amount of B is soluble in A, as we add more B, the stage comes when no more B dissolves and a second phase (P = 2) appears.
- Under this condition the most abundant phase will be A saturated with B (point a") and the minor phase will be B saturated with A (point a').
- Relative abundance's of the two phases are given by the lever rule.

$$n_\alpha l_\alpha = n_\beta l_\beta$$

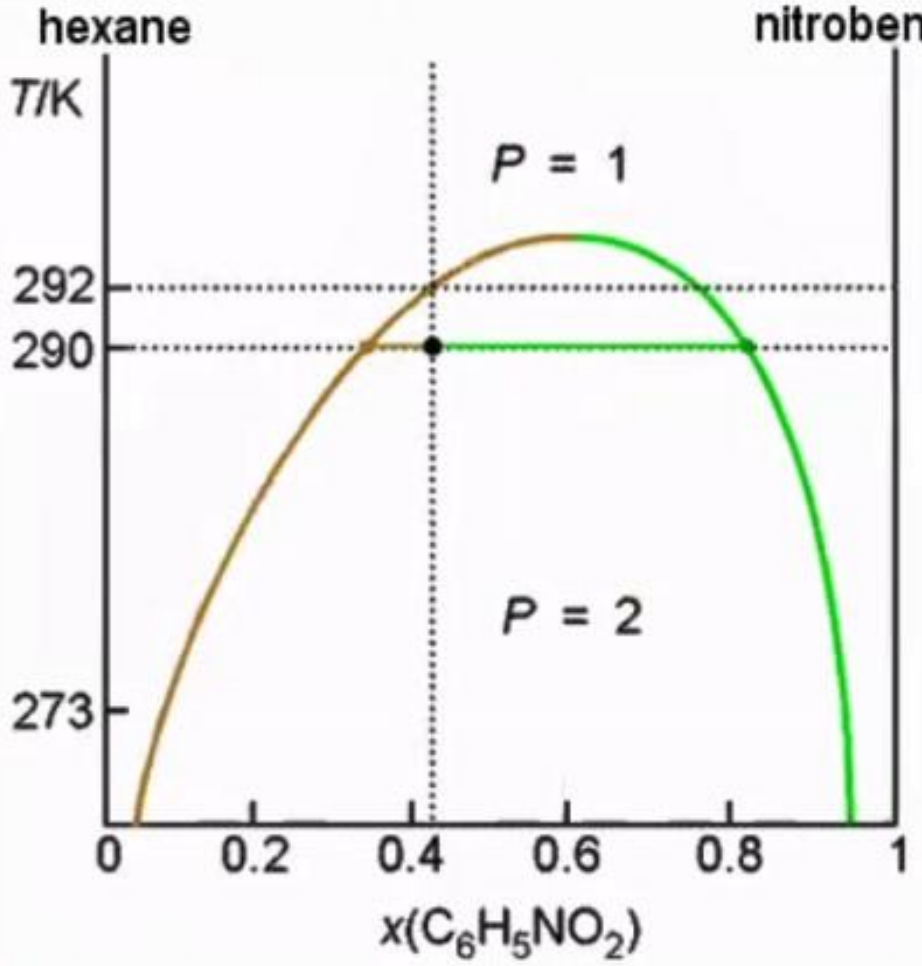
Phase separation

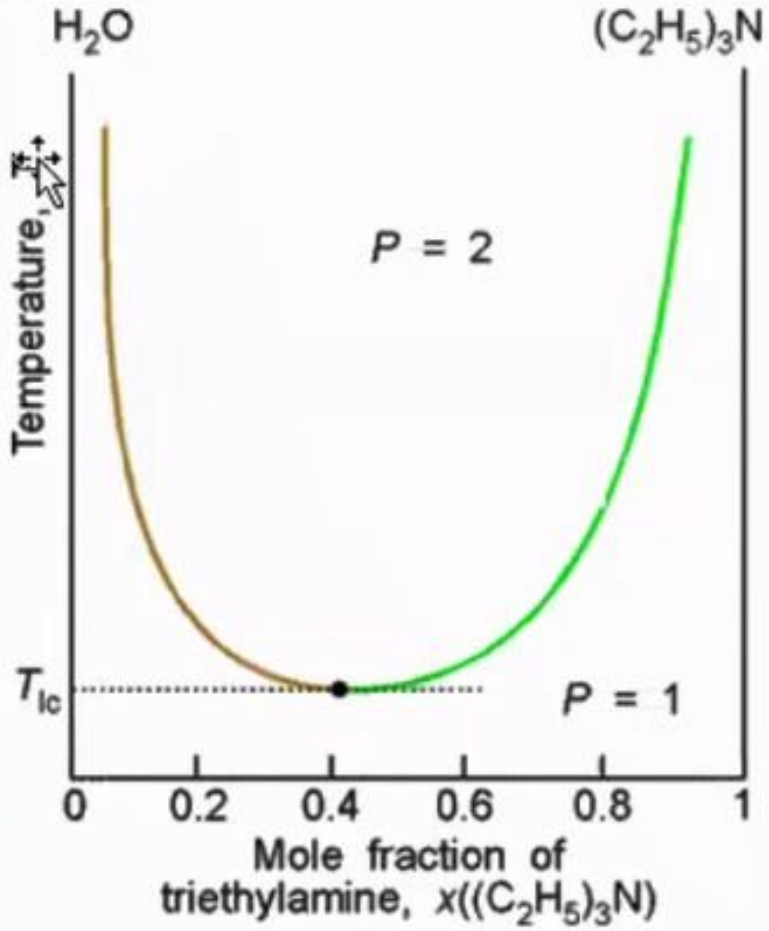


The temperature-composition diagram for hexane and nitrobenzene at 1atm. The region below the curve corresponds to the compositions and temperatures at which the liquids form two phases. The upper critical temperature, T_{uc}, is the temperature above which the two liquids are miscible in all proportions.

- There are some liquid pairs (e.g., triethylamine-water) for which mutual solubilities decrease with rise in temperature.
- As the temperature is decreased, the mutual solubilities increase and below the CST, the two liquids become miscible in all proportions.
- Such systems possess lower consolute temperatures.
- The variation of mutual solubility of triethylamine and water with temperature is shown below.
- Above 291.5 K, on shaking triethylamine and water, two layers are formed; but below 291.5 K, triethylamine and water are completely miscible with each other in all proportions.

Critical solution temperature



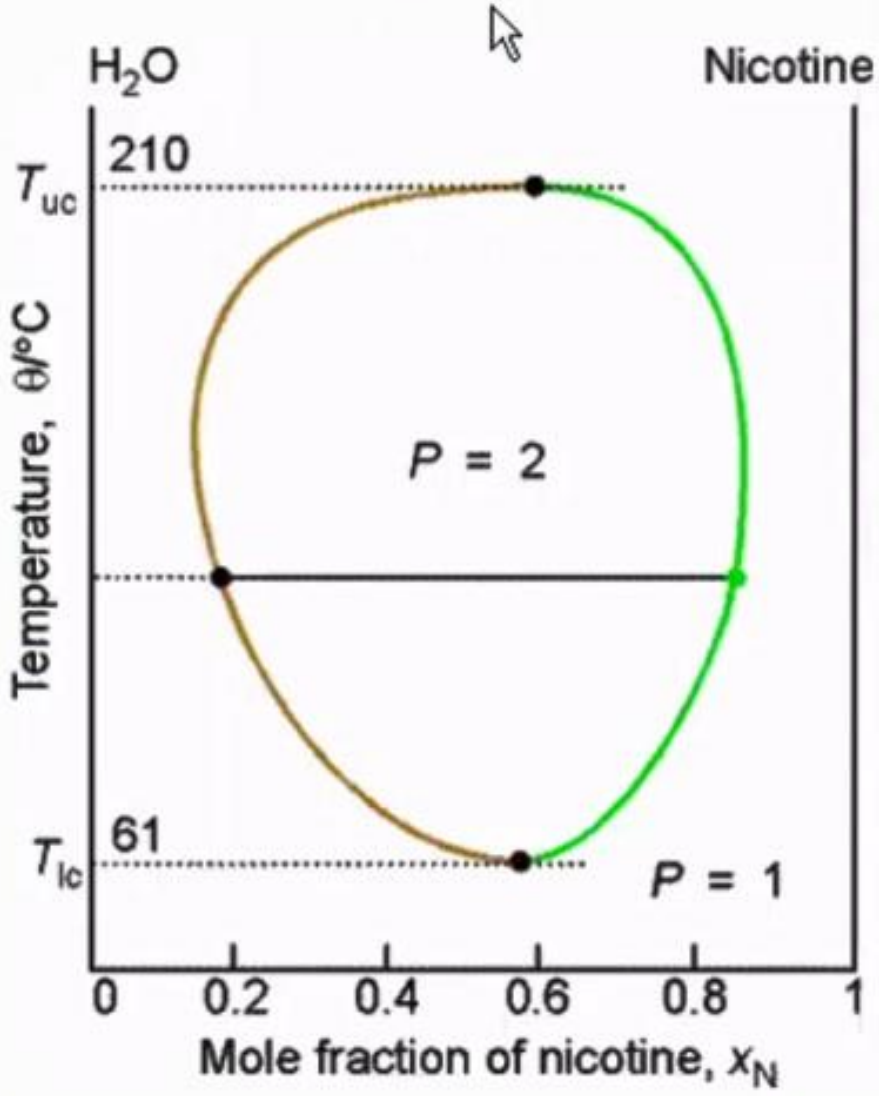


Critical solution temperature

- **The upper critical solution temperature (**Tuc) is the highest temperature at which phase separation occurs.
- An example is hexane and nitrobenzene. Above this temperature, the two components (hexane and nitrobenzene) are fully miscible.
- This exists because the greater thermal motion will overcome any potential energy advantage in molecules of one type being close together.
- Some systems show a **lower critical solution temperature (**Tlc) below which they mix in all proportions and above which they form two phases.
- An example is water and triethylamine. In this case at low T, they form a complex that breaks up at higher T.

- Some liquid pairs (e.g., nicotine - water) show both the upper and lower consolute temperatures.
- These liquid pairs are completely miscible above a certain temperature (upper consolute temperature, UCST) and below a certain temperature (lower consolute temperature, LCST).
- The variation of mutual solubilities of nicotine and water with temperature is shown below.
- Within the enclosed area, the liquids are only partially miscible, while outside the enclosed area, they are completely miscible.
- The composition & irresponding to both the upper and the lower consolute temperatures is the same, i.e., 34% nicotine.

Critical solution temperature



The temperature-composition diagram for water and nicotine, which has both upper and lower critical temperatures. Note the high temperatures for the liquid (especially the water): the diagram corresponds to a sample under pressure.

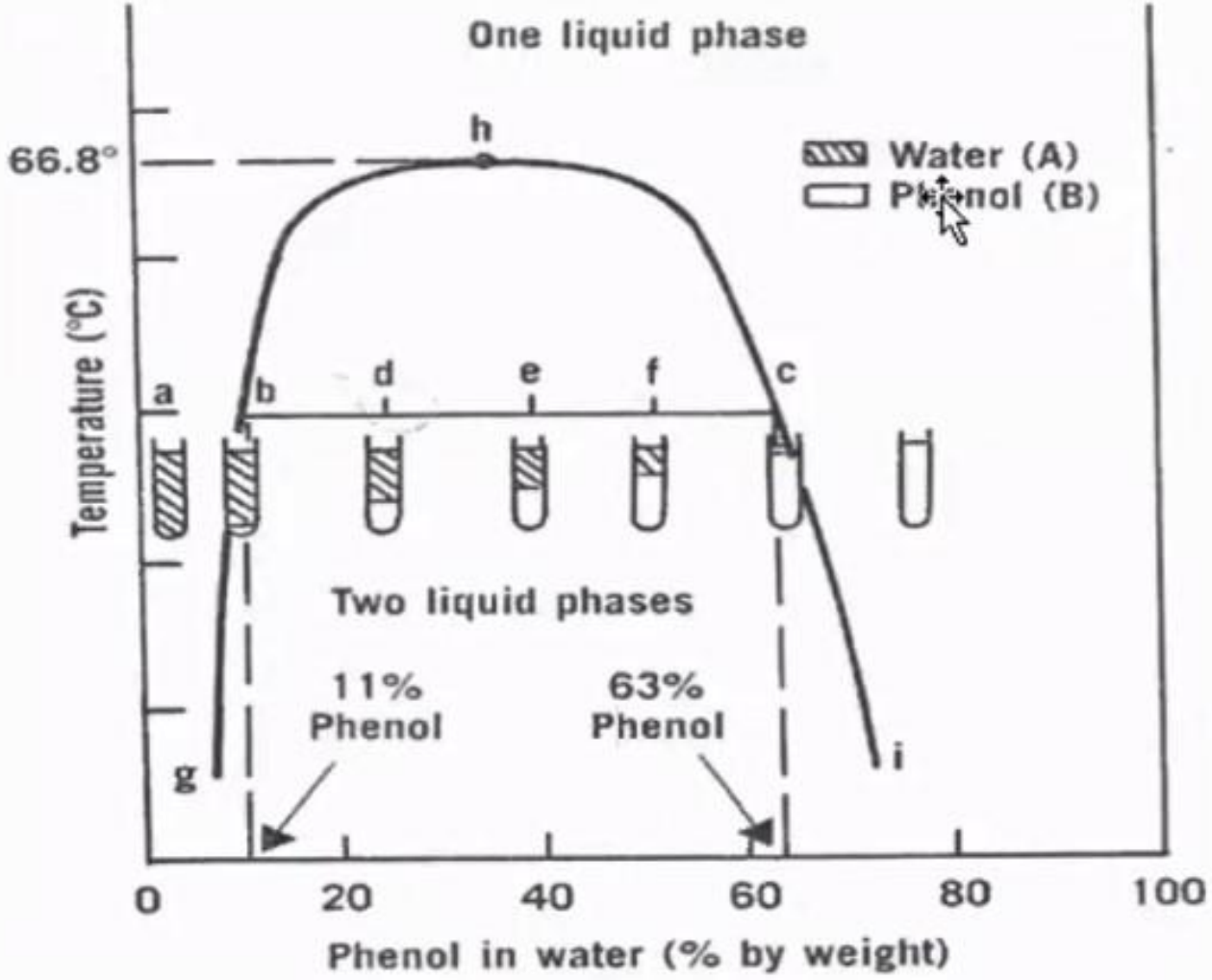
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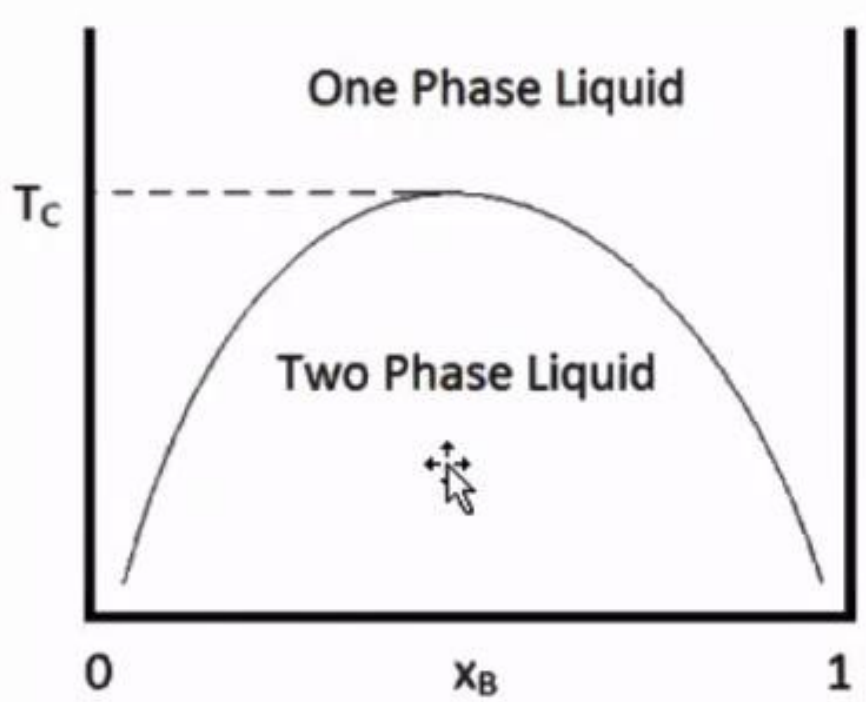
Partially miscible liquids

- Partially miscible liquid pairs are ether-water, phenol-water, methyl acetate- carbon disulfide, Trimethylamine- water, nicotine-water etc.
- When phenol and water are mixed, a certain amount of the two dissolves with the other due to hydrogen bonding and two conjugate layers of liquids are obtained.
- The upper layer of this system is water dissolved in phenol and the lower layer is phenol dissolved in water.
- The composition of these layers depends only on the temperature of the system and is independent of the quantities of the liquids mixed.

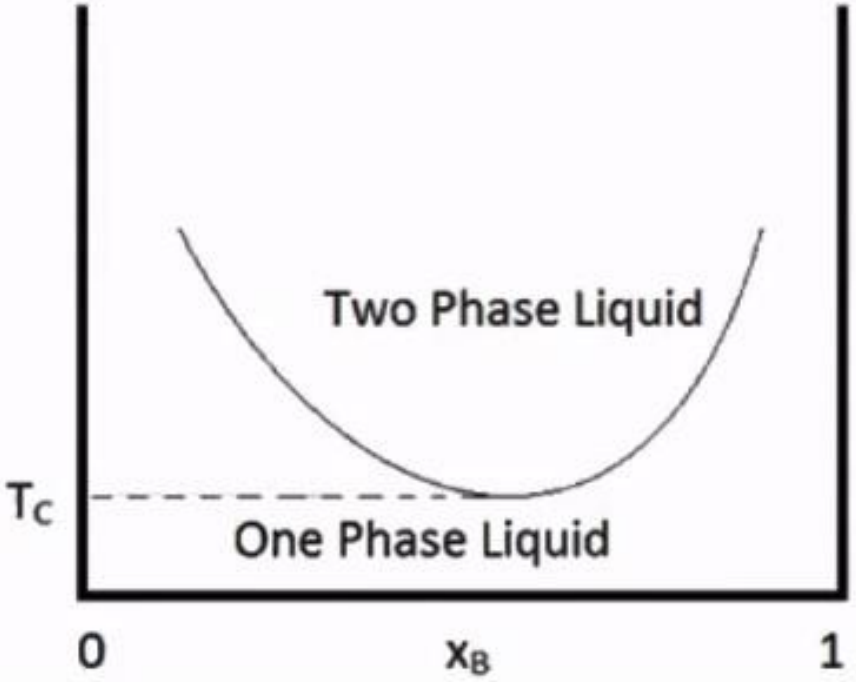
- For every temperature two solubilities can be obtained, one for phenol dissolved in water and the other for water dissolved in phenol.
- Generally, for partially miscible liquids solubility increases with increasing temperature and the temperature at which they are completely miscible is termed as the critical solution temperature (CST).
- The temperature above which the phases of a system are completely miscible is known as the upper critical solution temperature (UCST).

- Similarly, the temperature below which the phases of a system are completely miscible is known as the lower critical solution temperature (LCST).





Phenol-water



Trimethylamine- water

Role of added salts

- If the added salt dissolves in one of the phases only, it results in an increase in the UCST, and a decrease in the LCST, thus decreasing the miscibility of the system.
- Further, if the added salt is soluble in both the phases, then the UCST is lowered and the LCST is raised, thus increasing the miscibility of the system.
- This increase or decrease in the CST depends on the nature and the mass of the added substance and the composition of the system.
- It is observed that the UCST of phenol-water system increases with increase in concentrations of KCl impurities.

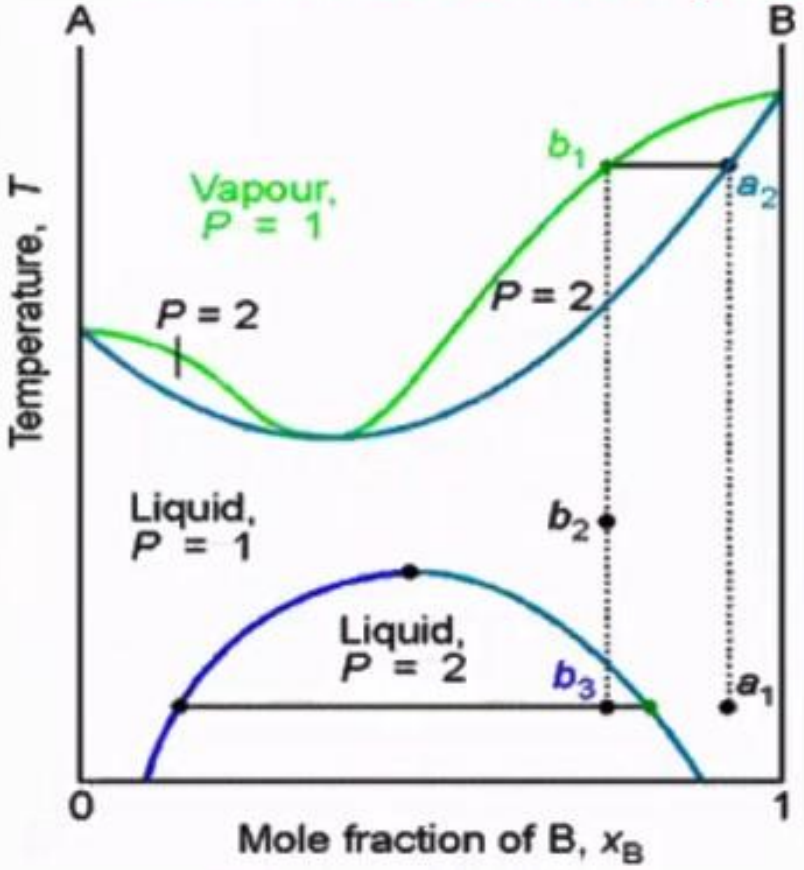
Role of added impurities

- The presence of an impurity, dissolved in one or both of the phases, changes the CST values as well as the liquid phase composition at CST.
- Substances soluble in only one of the liquids raise the upper CST and lower the lower CST.
- For example, 1% solution of NaCl raises the upper CST of phenol-water system by 12^0.
- About 0.12 molar solution of naphthalene (insoluble in water) in phenol raises the upper CST of phenol-water system by about 30^0.
- Substances which are soluble in both the liquids tend to lower the upper CST and raise the lower CST.
- For example, sodium oleate is soluble in both water and phenol. Addition of 1% solution of sodium oleate to phenol-water system lowers the CST value by 45^0.

Distillation of partially miscible liquids

- Consider a pair of liquids that are partially miscible and form a low boiling azeotrope (a common system, because both properties reflect the tendency of the two kinds of molecules to avoid each other).
- The figure shows the phase diagram of a system in which the liquids become fully miscible before they boil. Distillation of a mixture at a1leads to vapor with composition b_1, which condenses to completely miscible solution at b_2. Phase separation only occurs when the distillate is cooled to a point in the two-phase region such as point b_3.

Distillation of partially miscible liquids

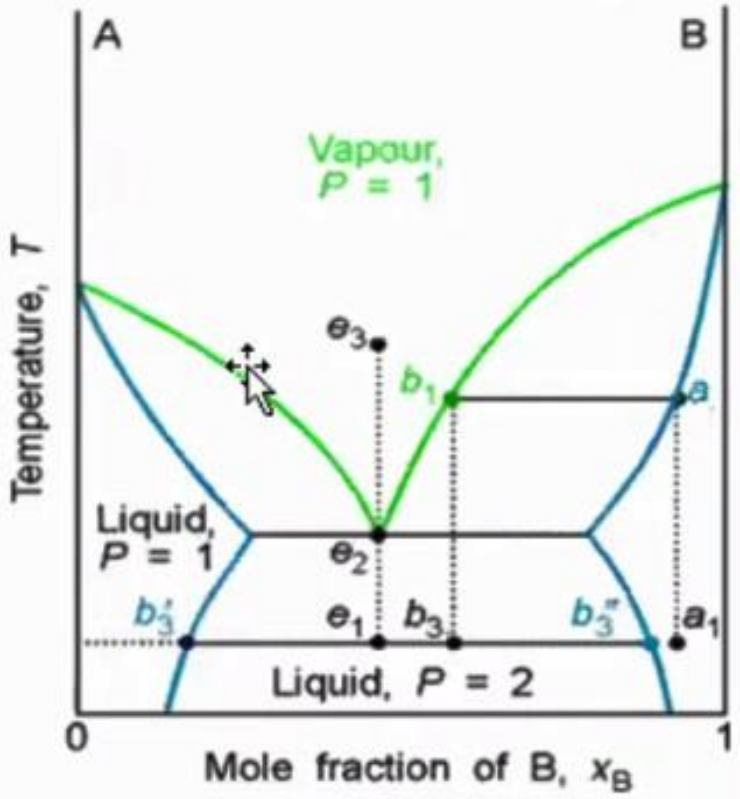


The temperature-composition diagram for a binary system in which the upper critical temperature is less than the boiling point at all compositions. The mixture forms a low-boiling azeotrope.

- This description only applies to the first drop of distillate. If distillation continues, the composition of the remaining liquid changes. In the end, when the whole sample has evaporated, the composition is back to a_1.

Distillation of partially miscible liquids

- This figure is for the situation in boiling occurs before complete miscibility. There is no upper CST.
- The distillate obtained from a liquid initially of composition a_1 has composition b_3 and is a two-phase mixture. One phase has composition b'_3 and the other has composition b''_3.

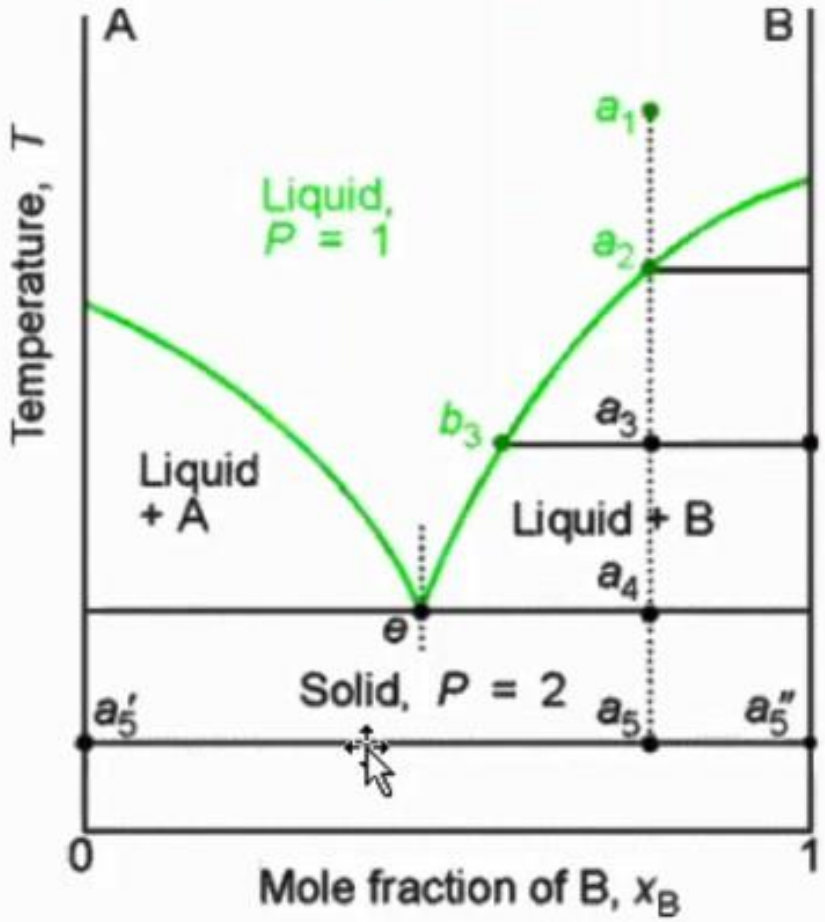


The temperature-composition diagram for a binary system in which boiling occurs before the two liquids are fully miscible.

Distillation of partially miscible liquids

- The behavior represented by isopleth *e* is interesting.
- A system at e_1, forms two phases, which persist (but with changing proportions) up to the boiling point e_2.
- The vapor of this mixture has the same composition as the liquid (azeotrope).
- Similarly, condensing a vapor of composition e_3, gives a two-phase liquid of the same overall composition.
- At fixed temperature, the mixture vaporizes and condenses like a single substance.

Liquid-solid phase diagrams



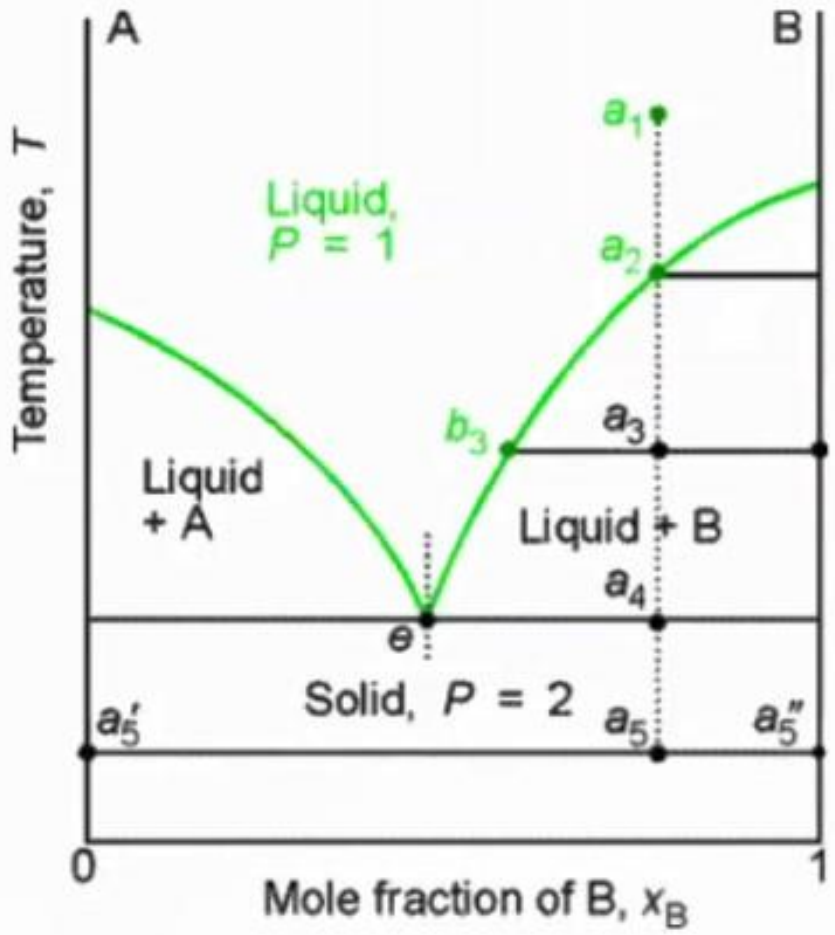
The temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids. Note the similarity to Fig. 8.26. The isopleth through *e* corresponds to the eutectic composition, the mixture with lowest melting point.

By Galib93

- Consider the two-component liquid of composition a1 in the diagram. The changes can be described as follows:

- 1) $a_1 \rightarrow a_2$: The system enters the two phase region labeled 'liquid + B'. Pure solid B begins to come out of solution and the remaining liquid becomes richer in A.
- 2) $a_2 \rightarrow a_3$: More of the solid forms, and the relative amounts of the solid and liquid (which are in equilibrium) are given by the lever rule. At this stage there are roughly equal amounts of each. The liquid phase is richer in A than before (composition given by b_3) because some B has been deposited.
- 3) $a_3 \rightarrow a_4$: At the end of this step, there is less liquid than at a_3, and its composition is given by *e*. This liquid now freezes to give a two-phase system of pure A and pure B.

Eutectics



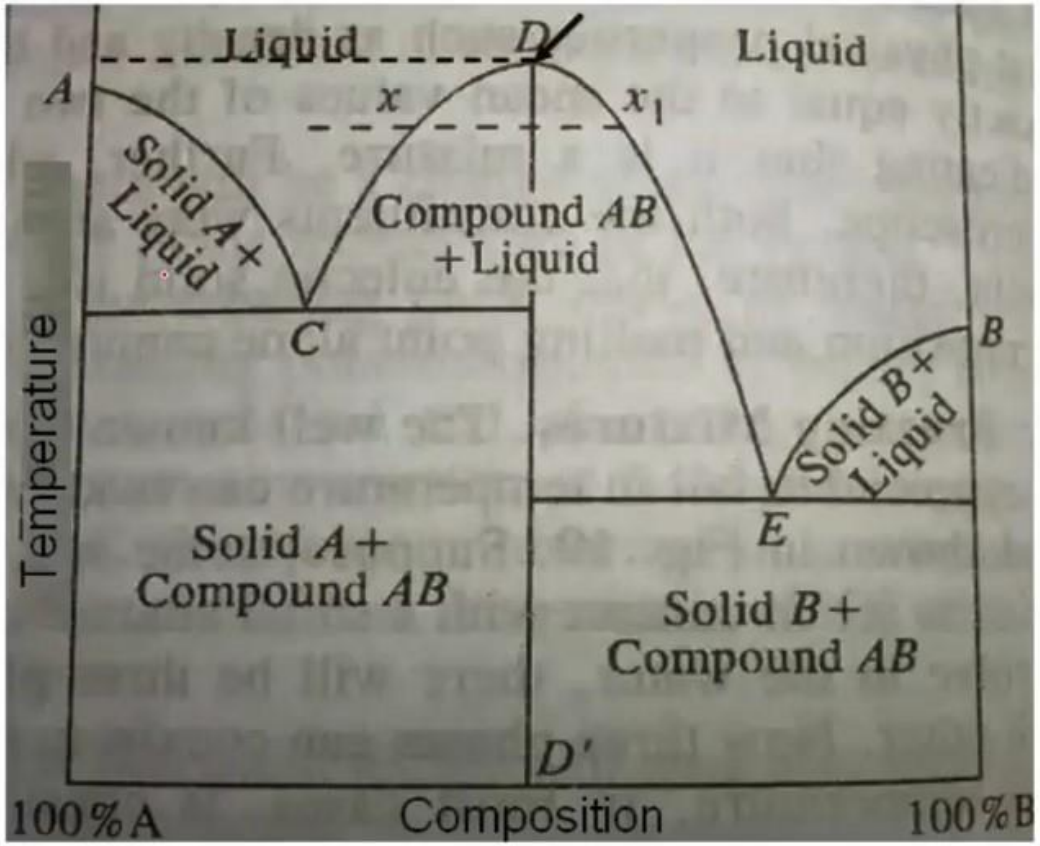
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Eutectics

- The isopleth at *e* corresponds to the **eutectic composition.**
- A liquid with the eutectic composition freezes at a single temperature, without previously depositing solid A or B. A solid with the eutectic composition, melts without change of composition at the lowest temperature of any mixture.
- Solutions with compositions to the right of e deposit B as they cool, and those to the left deposit A as they cool.
- Only the eutectic mixture (apart from pure A or solid B) solidifies at a single definite temperature (F' = 0 when C =2 and P = 3) without gradually unloading one or other of the components from the liquid.
- When a liquid reaches its eutectic composition, the temperature remains constant (F' = 0) until the whole sample solidifies. This is known as a **eutectic halt.**

Congruent Melting Point

- Congruent melting occurs during melting of a compound when the composition of the liquid that forms is the same as the composition of the solid.
- This generally happens in two-component systems.
- To take a general case, let A and B be the two components and AB a stable solid compound formed by their chemical combination.
- If we draw a phase diagram for the system, we notice that there are three solid phases, namely A, B and compound AB.
- Accordingly, there will be three fusion or freezing point curves AC, BE and CDE for the three solid phases.



- In the above phase diagram, we can notice that the top point D of the phase diagram is the congruent melting point of the compound AB because the solid and liquid phases now have the same composition.

- Evidently, at this temperature, the two-component system has become a one-component system because both solid and liquid phases contains only the compound AB.
- Congruent melting point represents a definite temperature just like the melting points of pure components.
- In the phase diagram, the congruent melting point D of compound AB lies above the melting points of pure components A and B.
- But it is not necessarily true. There are different types of systems known in which the congruent melting point is observed to be less than melting points of pure components.

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- When the solid compound *AB* and the liquid phase have identical composition at the maximum point on the freezing point curve, the corresponding temperature is said to be **the congruent melting point of the compound.**
- At point *D* since both the phases have the same composition, the system has one component only.
- Hence it is nonvariant.

$$F = C - P + 1 = 1 - 2 + 1 = 0$$

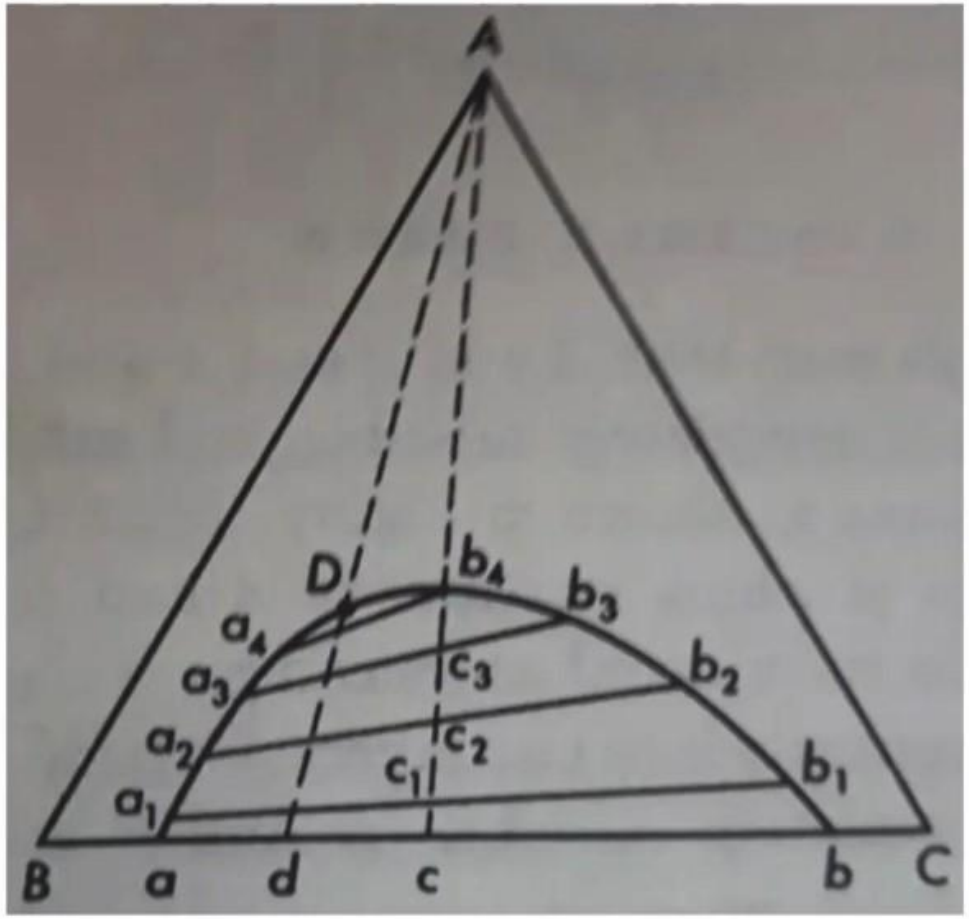
- The above diagram has the appearance of two simple eutectic diagrams joined together at the position of arrows.
- The diagram to the left represents the eutectic system *A*/*AB*, while that on the right the system *AB*/*B*.
- The phase diagram of the system zinc-magnesium is exactly of the type shown in figure.

Triangular phase diagram

- Consider a pair of liquids, B and C, that are partially soluble in each other at a given temperature and pressure.
- If we mix relative amounts of the two so as to exceed the mutual solubility limits, two layers will be obtained, one composed of a solution of B in C and the other of C in B.
- Now add a third liquid A which is completely miscible with both B and C.
- The increase of miscibility depends on amount of A added and on the amount of B and C present.
- If sufficient A is added, the two layers can be changed into a single solution composed of the three liquids.

- The changes in miscibility produced by progressive additions of A to mixtures of B and C can be followed on the diagram shown below:

- Points a and b designated the compositions of the two liquid layers resulting from mixing B and C in some arbitrary over-all proportion such as c, while line Ac shows the manner in which the over-all composition of c is changed by addition of A.



- When enough A is added to change c to c_1, the compositions of the two layers are shifted from a and b to a_1 and b_1 respectively.
- The line a_1b_1 through c_1 connects the compositions of the two layers in equilibrium, and is called tie line.
- In a similar manner the compositions are changes to a_2, a_3 and b_2, b_3 when the over-all compositions reach c_2 and c_3.
- Finally, at point b_4 sufficient A is added to form only a single layer of this composition, and thereafter only a single solution is obtained.
- The tie line for b_4 shows that the composition of the B-rich layer at the point of complete miscibility is not identical with b_4 but is equal to a_4.
- This fact indicates that complete miscibility is brought about not by coalescence of the two layers into one, but rather by the disappearance of the B-rich layer.

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- Complete miscibility by the merging of the two layers into one can occur only at one point on the diagram, D.
- At this point the compositions of the two layers become identical and the two solutions coalesce into a single liquid phase of the constant composition.
- Point D is called either the isothermal critical point of the system, or the plait point, and can be obtained only by adding A to a single mixture of B and C, namely, d.
- Any mixture of over-all composition outside this area will yield only a single homogeneous solution of the three liquids.

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- Curve aDb is frequently known as a binodal curve.
- Examples of systems of this type are acetic acid-chloroform-water (A-B-C) and acetone-water-phenol (A-B-C).